NNN-F-00141(DSA-DH)
7 December 1965

INTERIM FEDERAL SPECIFICATION

FLASK, SAYBOLT VISCOSITY

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 This specification covers one type of volumetric flask designed for measuring Saybolt viscosity of petroleum products.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

FEDERAL SPECIFICATIONS

DD-G-541	Glass (Laboratory).
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Setup.

(Activities outside the Federal Government may obtain copies of Federal Specification as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

FSC 6630

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D.C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

MILITARY SPECIFICATIONS

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

MILITARY STANDARDS

MIL-STD-105	Sampling Procedures and Tables for Inspection by attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of Military specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Designation D88-56	Standard Method of Test for Saybolt Viscosity.
ASTM Designation C148-65	Methods of Polariscopic Examination of Glass Containers.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103,)

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NATIONAL BUREAU OF STANDARDS

Circular 602

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402).

3. REQUIREMENTS

3.1 Material. The material shall be borosilicate glass conforming to DD-G-541, Type 1, Class A.

3.2 Construction. The flask shall be constructed in accordance with drawing requirements in ASTM Designation D88-56.

3.3 Marking. The flask shall be legibly and permanently marked with capacity, TC20° C, and manufacturer's name or registered trademark. The flask shall have a marking spot.

3.4 Annealing. Annealing of the flasks shall be such that ring strain shall be not greater than temper No. 1 when tested as specified in 4.4.2. Longitudinal strain, if present, shall be faint and highly diffused.

3.5 Workmanship. Workmanship shall be first class throughout. The flasks shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	2.5
Minor defects	II	4.0
For dimensional examination	S-2	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105. Unit of product shall be the flask.

Characteristic	Requirement	Test procedure	Inspection Level	ACL
Permanency of marking	3.3	4.4.3	S-2	*
Strain	3.4	4.4.2	S-4	*
Capacity	3.3	4.4.1	S-1	*

* Acceptance number shall be zero.

4.3 Examination. The flasks shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects.

Categories and defects*

Major
101 Flask not symmetrical
102 Constriction not symmetrical
103 Top not properly formed and reinforced
104 Marking spot missing, or not properly ground to take pencil mark
105 Ring around constriction not properly etched and filled
106 Flask not free from sharp edges
107 Flask not free from cracks
108 Flask not free from chips
109 Flask not free from protrusions

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Minor
201 Flask not free from bubbles
202 Flask not free from scratches
203 Flask not free from oil, grease, and extraneous matter

* Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The flasks shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Capacity. Capacity accuracy shall be determined in accordance with the method described in National Bureau of Standards Circular 602.

4.4.2 Annealing. Flasks shall be tested for strain in accordance with ASTM Designation C148-65.

4.4.3 Permanency of marking.

4.4.3.1 Resistance to alkali. The flask shall be immersed in 1 Normal sodium hydroxide solution at room temperature, covered, boiled for 1/2 hour, and allowed to stand in the alkali for an additional 1 hour. The flask shall then be removed, thoroughly rinsed with distilled water, and dried by rubbing vigorously with a cloth.

4.4.3.2 Resistance to acid.

4.4.3.2.1 Hydrochloric acid. The flasks shall be immersed in hydrochloric acid (specific gravity 1.18) at room temperature, covered, boiled for 1/2 hour, and allowed to stand in the acid for an additional 1 hour. The flask shall then be removed, thoroughly rinsed with distilled water, and dried by rubbing vigorously with a cloth.

4.4.3.2.2 Sulphuric acid. The flask shall be immersed in sulphuric acid (specific gravity 1.84) at room temperature and allowed to stand for 1 hour. The flask shall then be removed, thoroughly rinsed with distilled water, and dried by rubbing vigorously with a cloth.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One (1) flask, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table I shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table I. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	72 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a carton of appropriate size constructed in accordance with Federal Specification PPP-B-566, or PPP-B-676, except that commercial colors shall be acceptable, or PPP-B-636. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to the contents. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.3.2 Intermediate package. Intermediate package shall be a carton of appropriate size constructed in accordance with Federal Specification PPP-B-636. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

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5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4 of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic and method I closure shall be utilized.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

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5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Marking shall include the legend:

"GLASS - HANDLE WITH CARE."

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.

(b) Selection of applicable level of packing (see 5.2)

6.2 This specification does not include all types, classes, grades, sizes, etc. of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

Military custodians:
Army - MU
Navy - SH
Air Force - 67

Preparing activity:
Defense Supply Agency - DM

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDER
DEFENSE MEDICAL SUPPLY CENTER
3RD AVENUE & 29TH STREET
BROOKLYN, NEW YORK 11232

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PLATE NO. 15419 (BACK)

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863 WHICH IS OBSOLETE C-8279

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